

Review

Permafrost as a potential pathogen reservoir

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SUMMARY

The Arctic is currently warming at unprecedented rates because of global climate change, resulting in thawing of large tracts of permafrost soil. A great challenge is understanding the implications of permafrost thaw on human health and the environment. Permafrost is a reservoir of mostly uncharacterized microorganisms and viruses, many of which could be viable. Given our limited knowledge of permafrost-resident microbes, we also lack the basis to judge whether they pose risks to humans, animals, and plants. Here we delve into features of permafrost as a microbial habitat and discuss what is known about the potential for microbial pathogens to emerge in a warming climate as permafrost thaws. This review has broader implications for human health and ecosystem sustainability in the new Arctic environment that will emerge from a thawed permafrost landscape.

INTRODUCTION

Current predictions of permafrost land coverage range between 15% and 24% of the northern hemisphere.^{1,2} Permafrost soils are among the largest terrestrial carbon reservoirs on the planet, and in the northern hemisphere, permafrost soils are estimated to contain 1,014 petagrams carbon to a depth of 3 m.³ By definition, soils that have remained frozen for at least 2 consecutive years are accepted as permafrost, but most formations are much older than that and originate from prior glaciation periods. Frozen permafrost soils are overlain by an active layer that undergoes seasonal freeze-thaw cycles. Depending on when the soil froze, permafrost can contain organic depositions from prior periods of Earth's paleoclimate, including periods with a warmer and more carbon dioxide (CO₂)-rich atmosphere. Active-layer carbon stocks are newer deposits in comparison with permafrost.³

Permafrost is thawing because of anthropogenic climate change, and we can already observe large-scale changes in permafrost regions. Soil temperatures have increased by up to 2°C over the last decade, and the active layer has deepened significantly.⁴ Global warming has disproportionately warmed high-latitude and high-altitude permafrost (i.e., Tibetan plateau) areas, and these regions are projected to continue to experience greater temperature increases compared with other regions of the world.⁵ As a result, on a global scale, 22%–64% of permafrost is expected to thaw by 2100.⁶ Thawing permafrost creates a positive feedback loop by releasing nutrients such as water, allowing plants and microorganisms to grow. In low-drainage areas containing ice-rich permafrost or massive ground ice, gradual thaw has resulted in ground subsidence and inundation where formation of bogs, fens, and thermokarst (abrupt thaw) lakes mainly support microbial anaerobic fermentation and methane (CH₄) production.^{7,8} Thawed permafrost carbon stocks

are rapidly turned to CH₄, which further contributes to greenhouse gas accumulation.^{9,10} However, these observations do not reflect the entirety of the complex changes that Arctic landscapes are currently experiencing. Increasingly warmer and longer summer seasons rapidly melt ground ice where abrupt thaw intensifies subsidence and deepens the active layer.¹¹ Abrupt thaw in upland areas results in active-layer detachments or large-scale landslides and forms gullies and thaw slumps.¹² Model estimations predict abrupt thaw to cause carbon losses that are equivalent to 40% of those expected from gradual thaw.¹³ Currently we have a limited understanding of how microbes respond to such sudden changes and shifting quantities in ice, carbon, and nutrient deposits.⁴

Here we delve into features of permafrost that could make it a reservoir for microbes, including pathogens, and discuss what is known about the potential for microbial pathogens to emerge in a warming climate as permafrost thaws. In most cases, we find that the threats to public health are minor. We show that, although there is potential for bacterial human pathogens, such as *Bacillus anthracis* causing anthrax, to be problematic as permafrost thaws, there is less potential for other human pathogens. Most of the microorganisms and viruses that have been identified in permafrost to date target non-human hosts, such as plants and microorganisms. Plant pathogens could, however, pose a risk to native and introduced crops. Although fragments of human viruses have been recovered from remains preserved by permafrost, none have been shown to be infectious to humans. However, many of the microorganisms and viruses residing in permafrost have not yet been isolated or characterized. Therefore, their potential to pose future risks to humans, animals, and plants warrants consideration. These risks may increase with climate change as the landscape and vegetation changes and the human population becomes increasingly exposed to thawed permafrost lands.



MICROBES IN PERMAFROST

Permafrost soils have been shown to be microbially diverse terrestrial ecosystems.⁷ When frozen, many of the resident microorganisms in permafrost, unlike macroorganisms—think woolly mammoths—were able to adapt and survive in subzero temperatures.^{7,14} These psychrotolerant microorganisms¹⁵ also have to endure other stressful conditions, including high salinity, low water and nutrient availability,^{16–18} as well as anoxia and low pH in frozen bogs and fens.^{19,20} Permafrost “microbes” include bacteria, archaea, fungi, algae, protists, and viruses. Protozoa have also been isolated from permafrost.²¹ The types of microbes and their ability to survive depend on a variety of factors, including the permafrost age and chemical composition. Whether any of these microbes are potentially pathogenic to humans, other animals, or plants is a current concern and the topic of considerable debate.²²

Previous reviews have summarized the potential for pathogen emergence from other cryosphere environments, including glaciers and ice patches.^{23–25} For example, high frequencies of influenza A viral RNAs were detected in Siberian Lake ice and can potentially be transmitted via migratory birds.²⁶ In another study, one arthropod RNA virus and one plant DNA virus were recovered from ice patches in the Canadian Arctic.²⁷ Although outside of the scope of this review, which is focused on permafrost, these reports underscore the potential for the cryosphere to serve as a reservoir of emerging pathogens.

Bacteria and archaea in permafrost

It is known that the microbial communities that inhabit the active layer and permafrost soil layers are quite different in composition.⁷ Active-layer and permafrost microbes also have different vulnerabilities to increases in global temperatures. As permafrost thaws, the depth of the active layer increases. As a result, there is usually an increase in microbial activity in the newly formed active layer, which, in turn, intensifies greenhouse gas emissions⁷ and a transition in the composition of the microbial community.²⁸ Soil moisture content and overlying vegetation have been shown to be key drivers of the microbial response to permafrost thaw.^{20,29,30} In addition, differences in soil conditions (e.g., temperature, redox chemistry, and moisture content) are clearly of importance when determining the thaw response.²⁸

Permafrost soils have been shown to contain a wide range of soil bacteria, predominantly from the Terrabacteria supergroup, including *Actinobacteria*, *Firmicutes*, *Cyanobacteria*, *Chloroflexi*, *Bacteroidetes*, and *Proteobacteria*.⁷ Some representatives of less well known phyla have also been discovered in permafrost metagenomes, including metagenome-assembled genomes corresponding to *Candidatus* phylum Eremiobacteraeota and *Candidatus* phylum Dormibacteraeota in intact and thawing permafrost.³¹ Similar to bacteria, many different species of archaea have been found in permafrost, including representatives of *Euryarchaeota*, *Crenarchaeota*, and *Thaumarchaeota*. To date, the most commonly observed archaeal species in thawing permafrost are methanogenic Euryarchaeota, which can use fermentation byproducts to produce CH₄.³²

Conditions in permafrost can preserve nucleic acids and live cells that have mechanisms for survival under long-term freezing

conditions. Potential resources in permafrost, including organic carbon nutrients and water, are key to ensuring long-term survival. Because living organisms require water for survival, permafrost poses a unique challenge because of low water availability. Even a thin layer of salt brine can serve as a water reservoir because the salt increases the freezing temperature of water. Liquid water has been detected in permafrost at -10°C ,³³ and bacteria have been shown to be metabolically active at that temperature.³⁴

Increasing permafrost age also increases the necessity for microbial cells to cope with extended periods of stressful conditions.^{28,35} Mackelprang et al.²⁸ used 16S rRNA amplicon and metagenomic sequencing to profile the permafrost microbial communities existing in a permafrost chronosequence that ranged from 19,000 to 33,000 years ago (Pleistocene era).²⁸ They found that the age of the permafrost influenced the types of microbes present. For example, representatives of Firmicutes that can form resistant spores increased in relative abundance with age of the permafrost. Functional analysis of metagenomes revealed an enrichment in pathways for degradation of recalcitrant plant material and stress-related genes in older permafrost.³⁶

Most research has focused on understanding how interactions among post-thaw permafrost microbiomes are deterministic of observed carbon degradation rates and pathways. For example, recent evidence suggests that conditions mediated by thaw could give rise to acetoclastic methanogenesis, where hydrogenotrophic methanogenesis was predicted to be dominant in intact permafrost.³⁷ Drainage-driven dry soil conditions, on the other hand, can enhance aerobic processes and fuel CO₂ production, especially in uplands.¹⁰ Under these conditions, easily decomposable sugars, simple amino acids, and lipids are depleted because of oxidative decomposition, where hydrophobic carbon compounds are preserved in deep soils. In addition, multi-omics analyses of new thermokarst bog and fen formations showed a high abundance of methanogenesis-related genes, transcripts, and proteins, suggesting that CH₄ generation will increase in these systems when permafrost thaws.^{16,32} This has important implications for greenhouse gas emissions because CH₄ is a more potent greenhouse gas than CO₂. In contrast, little is known about the risk of pathogen exposure in thawed permafrost.

Viruses in permafrost

Viruses are the most numerically abundant biological entities on Earth,³⁸ and we have just begun to investigate the full extent of viral diversity in thawing permafrost soils. Recent surveys of the active layer of permafrost soils have identified that the vast majority of viruses are double-stranded DNA (dsDNA) bacteriophages (phages are viruses that only infect bacteria; bacteriophage in Figure 1). These phages are largely unrelated to other known phages, and a large portion of the community is active.³⁹ The phage community composition changes with permafrost thaw from soil-virus-like assemblages in the palsas with intact permafrost to a mixture of soil-like and aquatic-like in partially thawed bog to aquatic-virus-like assemblages in the fully thawed inundated fen habitats.^{40,41} Phage genomes are attuned to their environmental conditions by carrying auxiliary metabolic genes (AMGs) that can help their hosts overcome metabolic

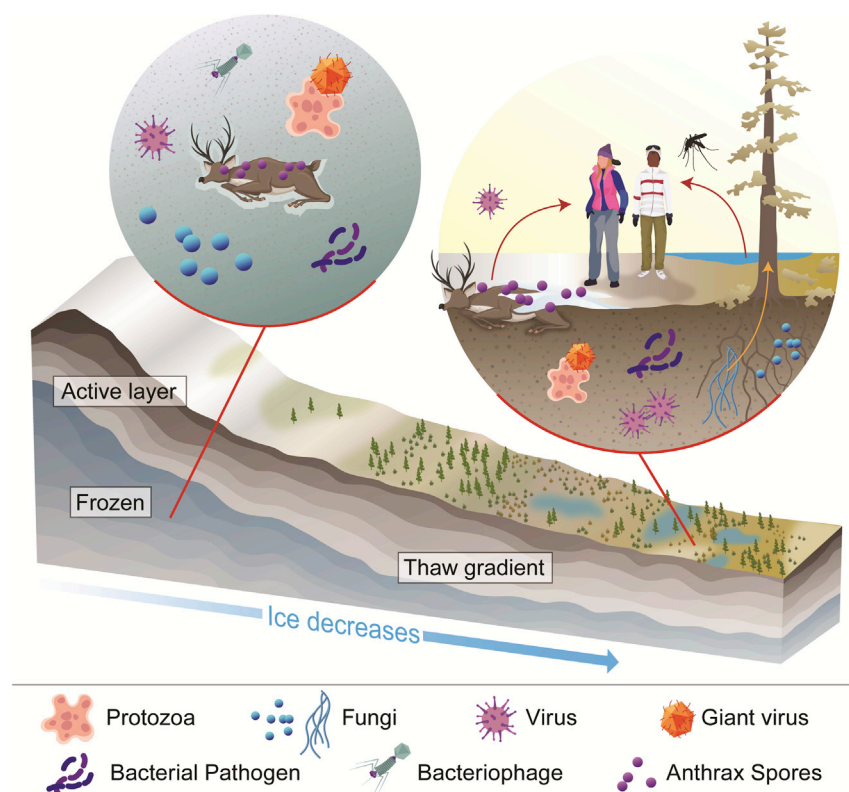


Figure 1. Emergence of potential pathogens along a permafrost thaw gradient

Left inset: examples of microbes in permafrost, including non-pathogenic (e.g., bacteriophage and giant viruses with bacteria and protozoa as respective hosts) and potential pathogenic microbes (e.g., spores of fungi, anthrax spores, other virus). Right inset: examples of pathogen emergence in thawed permafrost, including anthrax spores from animal carcasses and potential disease transmission vectors like mosquitoes.

Microbial activity in permafrost

Permafrost environments are subzero and often anoxic, which has led many to believe that resident microbes are not active. Over the last decade, there has been mounting research revealing diverse assemblages of active bacteria and viruses from permafrost environments.^{39,48,49} A multi-omics approach (i.e., metagenomics, metatranscriptomics, and metaproteomics) was used to determine which microbial populations were active along a permafrost thaw gradient.¹⁶ The proteome data revealed several proteins that were expressed by permafrost microorganisms to equip them to survive the harsh conditions in permafrost,

bottlenecks, such as resource limitations. For example, phages detected in thawing permafrost peatlands (soils with a large quantity of organic matter because of incomplete decomposition) have been found to carry an array of glycoside hydrolases that break down complex carbohydrates into labile carbohydrates, which, in turn, sustains a wide diversity of microorganisms.^{40,41}

Phages are the most intensely studied type of viruses in environmental systems, and, in contrast, our knowledge of single-stranded DNA (ssDNA) and RNA viruses is limited, especially in permafrost (virus in Figure 1). This is because (1) dsDNA phages are ubiquitous and abundant, (2) common methods to capture ssDNA or RNA viruses preclude quantitative ecological inferences⁴² or exclude them (e.g., metagenomes do not capture RNA viruses), and (3) most ssDNA and RNA viruses appear to infect eukaryotes.^{43,44} This suggests that we are missing an unknown quantity of environmental viruses that could be emerging pathogens.

Recently, a protocol was published to increase our detection and characterization of ssDNA viruses in soils underlain with permafrost.⁴⁵ Although ssDNA viruses were detected, they represented a small fraction of the viral diversity and abundance, and the potential hosts were bacteria or micro-eukaryotes (e.g., fungi and ameba).⁴⁵ Other studies of ssDNA⁴⁶ and RNA viruses^{29,47} in non-permafrost soils showed that they are abundant and diverse groups of viruses with clear differences in bulk soil versus rhizosphere communities and also identified bacteria or micro-eukaryotes as potential hosts. However, we have yet to fully understand the full extent of global viral diversity, including in permafrost soils.^{29,46,47}

including cold shock proteins. The metatranscriptome data also revealed actively transcribing bacterial populations in permafrost, including representatives of Firmicutes, Bacteroidetes, Acidobacteria, and Proteobacteria phyla.¹⁶ It is noteworthy that the active microbes detected in permafrost in the study by Hultman et al.¹⁶ were representatives of “typical” environmental bacteria, not pathogens.

With respect to viruses, there is a prevailing reductionist point of view that claims viruses are not truly active but, rather, that they are like enzymes, where random collisions lead to a series of chemical reactions. This is because without cells, viruses cannot multiply, and they do not have the ability to generate ATP. Therefore, to determine whether a virus is active, evidence is needed that the virus is able to infect a host and cause a change in the host’s ecology. In previous work, viruses have been co-cultured with putative hosts to determine activity, but determining the host and culturing a virus can be difficult and time consuming.⁵⁰ Several advances have recently been developed to determine activity of viruses in environmental samples. A few examples include (1) bioorthogonal noncanonical amino acid tagging (BONCAT), a technique that adds a synthetic amino acid to a sample that allows tracking and quantification of newly synthesized proteins;⁵¹ (2) stable isotope probing (SIP), which uses an isotope with more mass to label newly synthesized DNA;³⁹ (3) and multi-omics (i.e., metagenomics, metatranscriptomics, and metaproteomics²⁹), such as described above,¹⁶ which can reveal viral presence, gene expression, and protein production in a sample. Unfortunately, viruses can have several infection styles, each having variable effects⁴² (warranting further investigation). Although these techniques have been

applied to an array of habitats, few have been applied to determine viral activity in permafrost. Recently, SIP-metagenomics was used to determine viral activity in bog samples³⁹ that were incubated for 6–12 months under conditions simulating the freezing winter temperatures in the presence of H₂¹⁸O. Active populations incorporated ¹⁸O into their DNA when replicating and could thus be distinguished from inactive populations. This study revealed that only a subset of bacteria was active under the simulated winter conditions and suggested that they survived by fermentation and carbon-scavenging processes. However, the majority of the detected phages were active and linked to the active bacterial populations, highlighting their importance in structuring of the microbial community. This study focused on active phages that have prokaryotic hosts. Thus, the activity of potential plant, animal, or human viruses was not determined.

PATHOGENS IN PERMAFROST

An estimated five million people currently live in Arctic regions that have underlying permafrost lands.⁵² This number is even higher when including tourism and trade in sub-Arctic permafrost areas.^{53,54} As permafrost thaws, there is thus a potential for emerging pathogens to infect humans. Building on current knowledge about microbial life and activity in permafrost in general, we will focus on what has been reported about potential pathogens and exposure from thawed permafrost. We will highlight examples of bacterial, fungal, and viral pathogens in permafrost that can pose future risks to humans, other animals, and plants (see the details of each in Table 1). Our survey of the literature will conclude that we still have much to learn about potential pathogen risks from thawing permafrost.

Potential bacterial pathogens in permafrost

Bacterial pathogens may be preserved in a quiescent state in permafrost and become activated when the permafrost thaws and conditions are favorable for growth (bacterial pathogen in Figure 1). One of the best-known examples is that of *Bacillus anthracis*, the agent of anthrax disease that is globally distributed with potential risks to humans, livestock, and wildlife.⁵⁵ Spores, such as those from *B. anthracis*, are notoriously robust during long-term storage under freezing conditions (anthrax spores in Figure 1). In 2016, an outbreak of anthrax was reported in Siberia as a result of exposure to infected carcasses that had previously been frozen in permafrost. As the permafrost thawed, the carcasses became exposed, and anthrax spores were released. As a result, one person and over 2,000 reindeer died after ingestion of the spores.^{56,57} Anthrax outbreaks have also occurred when infected cattle that were buried were re-exposed after permafrost thaw.⁵⁸

Stella et al.⁵⁹ modeled the risk of future anthrax exposure and found that the annual risk of anthrax infection is strongly correlated with active-layer depth. The model distinguished between freshly released spores from thawed carcasses and those that become reactivated as a result of permafrost thaw. Seasonal forcing by thawing and herding was found to favor endemic disease transmission. Several knowledge gaps exist that are needed to improve the model, including how spores are vertically distributed in permafrost soils. The results from Stella et al.⁵⁹ suggest that seasonal thawing of permafrost,

together with longer periods of warmer temperatures, could result in an increased prevalence of re-emergence of anthrax spores and increased risk of associated disease in the Arctic. A compounding reason for anthrax outbreaks is an increase in reindeer with a lack of immunity⁶⁰ that could contribute to the spread and infectious capacity of the pathogens. Therefore, one potential mitigation strategy to help control future outbreaks would be to increase reindeer vaccinations.

There are several reports about prevalence of antibiotic resistance genes in permafrost bacteria^{61–63} that could make known pathogens more problematic. Antibiotic resistance poses a threat to human health by reducing the efficacy of antibiotic treatments. Antibiotic resistance in pathogenic bacteria is assumed to be a modern phenomenon. However, targeted metagenomic analyses from 30,000-year-old Alaskan permafrost showed resistance to β -lactam, tetracycline, and glycopeptide, predating the discovery and use of medicinal antibiotics.⁶⁴ This finding suggests that permafrost can serve as a reservoir of antibiotic resistance genes. Prevalence of antibiotic resistance in permafrost bacteria is likely an emergent property for competitive fitness rather than pathogenicity. As permafrost thaws, there is the potential for an increase in bacterial populations that bear plasmids with antibiotic resistance genes that have the potential to spread. However, this potential concern is likely no greater than that posed by non-permafrost soils that also harbor abundant antibiotic resistance genes.

Potential fungal pathogens in permafrost

Sequencing has revealed that several fungi are present in permafrost, including species of *Geomyces*, *Cladosporium*, *Aspergillus*, and *Penicillium*.⁶⁵ Some fungi that have been isolated from permafrost are potential plant pathogens (fungi in Figure 1). Ancient permafrost from Siberia (16,000–32,000 years old) was found to contain several fungal operational taxonomic units (OTUs) from three phyla: Ascomycetes, Basidiomycetes, and Mucormycota.⁶⁶ This study found that approximately one-third of the species detected were representative of plant-associated fungi, including pathogens. Within thermokarst bogs, the plant pathogens *Galerina paludosa* and *Hyaloscypha* increased in relative abundance after permafrost thaw.⁶⁷ The fungi were likely trapped together with plant material as the permafrost froze. Most fungal isolates from permafrost form small conidia, suggesting that this is a survival mechanism for long-term incubation in the cryosphere. Whether the fungi were still alive is not known. However, several of the fungal taxa were designated to be putative psychrotrophic fungi with representatives known to produce dark pigments and spores to help with survival under stress conditions.

Representatives of fungal insect pathogens have also been detected in ancient Siberian permafrost.^{66,68} Some taxa were putative colonizers of skin or dung of mammals, potentially reflecting the presence of a range of insect and other animal species that thrived at the time (Pleistocene to early Holocene eras). Lydolph et al.⁶⁸ studied ancient DNA from a permafrost tunnel in Siberia that ranged from the modern age to 400,000 years ago. They also found ancient DNA with fungal representatives of coprophilic and keratophilic fungi as well as plant pathogens. The main representatives were from the Ascomycetes, Basidiomycetes, and Zygomycetes phyla. However, to our knowledge,

Table 1. Examples of potential pathogens in permafrost and associated ice patches

Biological entity	Name	Disease caused	Type of pathogen	Sample detected	Outbreak	Reference
Bacterium	<i>Bacillus anthracis</i>	anthrax	human or animal pathogen	Siberia permafrost	In 2016, a person and thousands of reindeer died; between 1897 and 1925, 1.5 million deer died.	56–58
DNA virus	Variola virus	smallpox	human or animal pathogen	northeastern Siberia	N/A	81,82
	ancient caribou feces-associated virus (aCFV)	asymptomatic infection	plant pathogen fungal pathogen	700-year-old caribou feces from a subarctic ice patch ^a	N/A	27
Fungus	members of the genera <i>Ascochyta</i> , <i>Entyloma</i> , <i>Epicoccum</i> , <i>Fusarium</i> , <i>Gaeumannomyces</i> , <i>Microdochium</i> , <i>Phaeosphaeria</i> , and <i>Phoma</i>	leaf spot, defoliation, and a reduction in yield and crop value	plant pathogen	ancient permafrost from Siberia (16,000–32,000 years old)	N/A	66
	members of <i>Cordyceps</i> and <i>Paecilomyces</i>	paralysis or mortality	insect pathogen	ancient permafrost from Siberia (16,000–32,000 years old)	N/A	66,68
RNA virus	Cripavirus	paralysis or mortality	insect pathogen	700-year-old caribou feces in the subarctic	N/A	27
	H1N1 virus	influenza	human or animal pathogen	frozen lung tissues of 1918 influenza victims	N/A	85
	tomato mosaic Tobamovirus (ToMV)	tomato mosaic disease	plant pathogen	140,000-year-old Greenland ice core ^a	N/A	88

^aNon-permafrost cryosphere samples.

no active fungal pathogens in permafrost that directly infect humans have been reported to date.

Potential viral pathogens in permafrost

It is currently unknown how many viruses are trapped in permafrost, but when thawed, so-called “sleeping” or “zombie” viruses may become free to infect hosts. Although viruses can be preserved in a frozen environment for long periods of time,⁶⁹ endurance of environment pressures during thawing without bursting is critical for viruses to remain intact and infectious. One example of a class of viruses that have managed to survive in ancient permafrost are giant viruses. Giant viruses or nucleocytoplasmic large DNA viruses (NCDLVs) are known to have durable multiple layers enclosing virion nucleoids, making them resistant to the harsh permafrost environment⁷⁰ (giant viruses in Figure 1). Two members of giant viruses, *Pithovirus sibericum*⁵⁰ and *Mollivirus sibericum*,⁷¹ have been isolated and revived from 30,000-year-old Siberian permafrost. Although these viruses were preserved frozen in permafrost for at least ~30,000 years, after thawing, they were able to infect a modern-day version of their host.⁵⁰ The isolated giant viruses were reported to have thick outer walls of 40–70 nm,^{71,72} comparable with 20–80 nm for Gram-positive bacteria.⁷³ However, the giant viruses that have been characterized to date do not pose a threat to humans and other animals because their natural hosts are micro-eukaryotes, such as protists and algae (protozoa in Figure 1). Another giant virus, Mimivirus, which was first isolated from water samples for investigating a pneumonia outbreak,⁷⁴ was found to be capable of entering human and mouse myeloid cells, resulting in a significant increase in viral DNA and a decrease in macrophage viability.⁷⁵ A clinical study also reported a high prevalence of Mimivirus antibodies in individuals with pneumonia with unknown underlying mechanisms.⁷⁶ Although viruses often have a narrow and specific host range, these findings challenge the assumption that micro-eukaryotes are exclusive hosts for giant viruses, highlighting a major knowledge gap in virus-host relationships. Currently, however, there is no evidence showing that isolated and revived giant viruses from permafrost can cause human infections.

The majority of DNA viruses in permafrost environments are phages and, therefore, cannot cause disease in humans. In fact, phages are arguably the most well-studied type of virus, and the majority of them (and viruses in general) are host specific. This means that they can only infect bacteria that are of the same species and often only the same strain.^{77,78} Despite phages being largely bacterial host specific, they appear to be major community-structuring agents that uphold a diverse permafrost microbiome. A diverse microbiome prevents any specific bacterium (e.g., a human pathogen) from growing in abundance. This means that, beyond phages killing pathogenic bacteria, thereby protecting us, they can create an ecosystem where pathogenic bacteria are less likely to be of concern to humans.⁷⁹ Given current evidence of the widespread presence of phages in all soils, including permafrost, they are very unlikely to represent a threat to humans.

Other types of DNA viruses, such as Variola virus, which causes smallpox, that infect humans may be present in permafrost environments, but they have not been detected in sufficient numbers or with good enough quality to pose a threat.⁸⁰

Because DNA viruses co-evolve and replicate in the presence of their hosts, for pathogenic human viruses to be present in sufficient quantities to cause infection, human/animal remains would have to be preserved in a condition that allows viral survival. However, this is highly unlikely. For example, Variola virus DNA was recovered from several archeological sites in northeastern Siberia dating back to the late 17th–19th century. Smallpox victims mummified in the permafrost were found, but the Variola virus DNA was highly degraded, and, thus, the virus was not infectious.^{81,82} Therefore, although human DNA viral pathogens can be detected in permafrost, the risk of infection is negligible and only of concern when the virus is intact and present in sufficient quantities to cause disease. It is noteworthy that, to date, there have been no reported cases of viral infections from thawed permafrost in humans.

To our knowledge, there is no genetically informed ecological study of the RNA viral community in permafrost. One reason why RNA viruses are understudied compared with DNA viruses is that RNA is less stable compared with DNA⁸³ and more prone to degradation during the extraction process. One example of an RNA virus is the H1N1 influenza A virus that caused the 1918 pandemic. This virus killed an estimated 50 million people, which overwhelmed the burial of infected humans, leading to mass graves and unburied corpses.⁸⁴ These mass graves largely went unmarked, leaving the potential for humans to interact with these corpses should permafrost thaw. In the 1990s, attempts were made to reconstruct or revive the 1918 pandemic influenza virus from frozen samples, but they were unsuccessful because the RNA was highly degraded. In the early 2000s, advances in technology allowed successful reconstruction of the H1N1 virus from permafrost, but it was not viable.⁸⁰ Complete hemagglutinin (HA) genes, encoding binding proteins to initiate infection, were detected in reserve-transcribed RNAs extracted from frozen lung tissues of 1918 influenza victims who were buried in a mass grave in permafrost.⁸⁵ Although this last example could be taken as a cause for concern, the viruses were not infectious. Ancient RNA extracted from wolf tissue buried in late Pleistocene permafrost showed no traces of RNA viruses.⁸⁶ To date, there is no report of intact and infectious RNA viruses directly isolated from permafrost. Therefore, although RNA viruses can be preserved in permafrost, based on our current knowledge, the risk of these RNA viruses being infectious to humans or other animals is unlikely.

Permafrost thaw is causing major changes in plant composition and productivity and will undoubtedly affect plant viral pathogens.⁸⁷ Previously, plant viral pathogens, such as ancient caribou feces-associated virus (aCFV) and tomato mosaic Tobamovirus (ToMV) were recovered from ice.^{27,88} aCFV is a DNA virus that can infect *Nicotiana benthamiana* (a close relative of tobacco) and may even infect other plants or fungi (e.g., mosses and lichens); it was recovered from caribou feces dating back 700 years before present (BP).²⁷ To test whether aCFV was still infectious, a modern version of *N. benthamiana* was exposed and successfully infected. Although the plant became infected, it had an asymptomatic infection, likely because of changes in the plant species over time. Unlike aCFV, ToMV is one of the most common plant viruses because it has the ability to infect many different plants, such as cucumber, lettuce, beet, and pepper.⁸⁹ This RNA virus is very stable, having been shown previously to

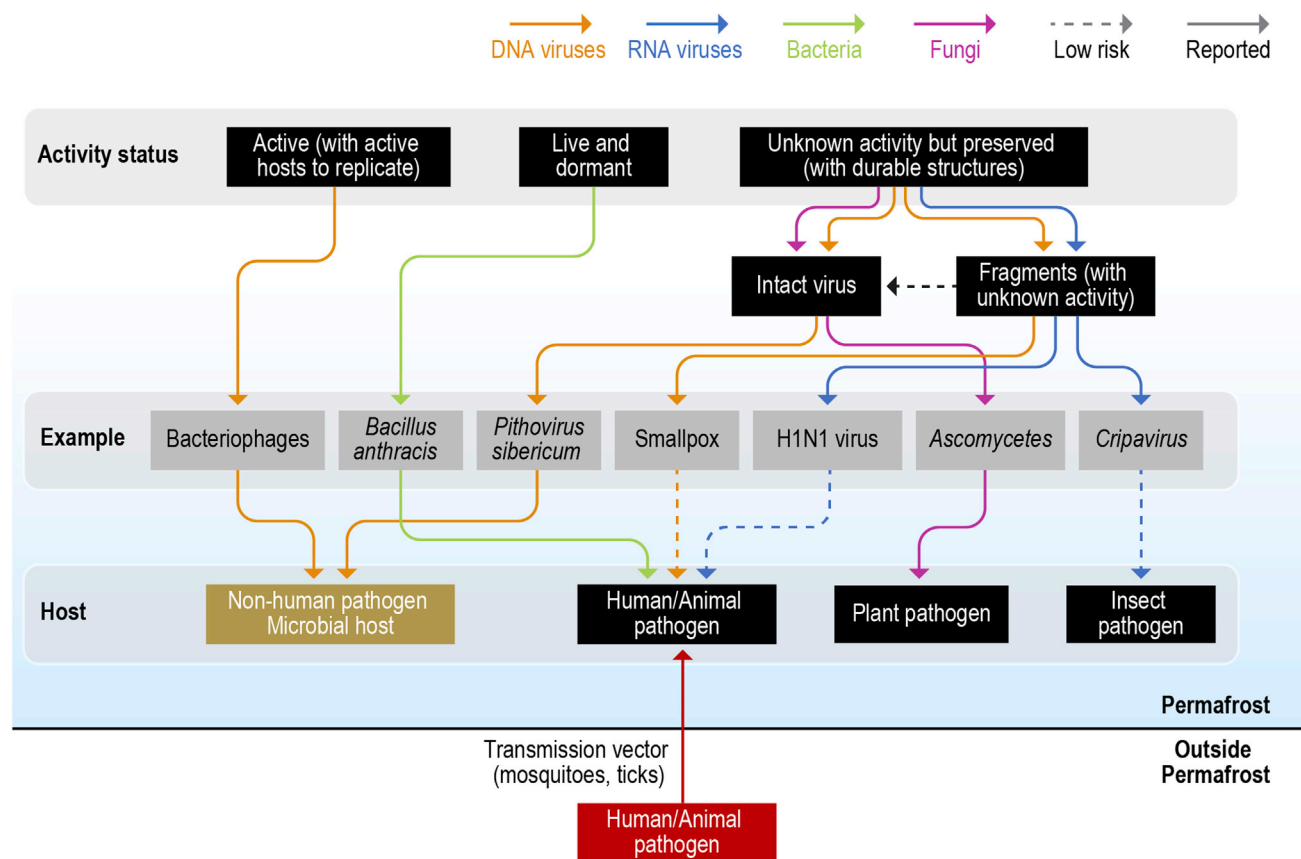


Figure 2. Microbes in permafrost can infect a diverse range of hosts

Microbes in permafrost are categorized into three activity statuses (active, live and dormant, and unknown activity but preserved) in intact or fragmented forms. Examples of microbes (DNA viruses, orange lines; RNA viruses, blue lines; bacteria, green lines; fungi, pink lines) are listed that potentially infect bacteria, protozoa, humans/animals, plants, and insects. Low-potential pathogenicity cases are indicated by dashed lines. Reported pathogenicity cases are indicated by solid lines.

survive UV radiation and desiccation, and was recovered from ice cores dating back up to 140,000 years ago.⁸⁸ It is unknown whether the recovered ToMV could still be infectious, but it presents a potential problem because it is one of the most destructive diseases, resulting in serious crop losses.⁹⁰ Introduction of non-native crops to soils, such as recently thawed permafrost soils, could also increase the likelihood of a viral plant epidemic.^{87,90}

CONCLUSIONS AND PERSPECTIVE

Although we have focused on permafrost as a potential reservoir of pathogens, there are other factors to consider in a warming Arctic climate. As we have summarized in this review, although some of the microorganisms and viruses that are preserved in permafrost can be active after thawing, the risks to human health are generally low (Figure 2). In the age of coronavirus disease 2019 (COVID-19), the human population has become painfully aware of the threat of emerging pathogens from “wild” spaces. Although speculative, there is still a risk that an unknown pathogen could arise from “wild” permafrost where it has survived over a long period of time in a dormant or quiescent state. A more immediate concern is expansion of modern infectious

diseases via vectors (e.g., ticks and mosquitoes) in areas where permafrost has thawed and the soil becomes wetter, or collapses into thermokarst bogs, as the climate warms.^{91–93} Increased temperature and precipitation can expand the geographical distribution of animal vectors.⁹⁴ Many of these vectors have been reported to carry human pathogens such as *Francisella tularensis*,⁹⁵ West Nile virus,⁹⁶ and Zika virus.⁹⁷ Transmission vectors can facilitate the spread of permafrost bacterial pathogens. For instance, anthrax has been reported to be carried by mosquitos (*Aedes aegypti* and *Aedes taeniorhynchus*) and flies (*Stomoxys calcitrans*).⁹⁸ Another pending concern is that increased temperature is correlated with an increase in food- and waterborne diseases.⁹⁹ Accordingly, there have been recent reports of several enteric disease outbreaks in the Arctic that were caused by bacterial pathogens, including *Campylobacter*, *Escherichia coli* 0157:H7, *Vibrio parahaemolyticus*, and *Salmonella*. Other vector-borne diseases (e.g., by rodents) that could increase in frequency in a warmer Arctic environment include tick-borne encephalitis (TBE), nephropathia epidemica, tularemia, borreliosis, malaria, toxoplasmosis, and Sindbis virus infection.²² Finally, airborne diseases, including viral respiratory tract infections and influenza, have been correlated with shifts in temperature and humidity. However, the

research underlying increased disease prevalence in the Arctic is still scarce. Therefore, we conclude with a call to the broader research community to increase research efforts in this area, as recently highlighted in a general call to increase research on the effect of climate change on microorganisms in the environment.¹⁰⁰

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AUTHOR CONTRIBUTIONS

J.K.J. coordinated the scope and content of the review. N.T., R.W., and G.T. drafted Figure 1. R.W. drafted Figure 2. All authors contributed equally to the writing of this review.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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